Recent Advances in Plasmonics and Nanophotonic Realizations

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Abstract
Plasmonics has undergone substantial evolution over the past decade, broadening from the study of passive metallic nanostructures supporting surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPR) into a multidisciplinary field that intersects with quantum information, solar energy conversion, advanced sensing, and nanofabrication. This review critically synthesises key developments across nine interconnected frontiers: (i) the shift from passive to active and reconfigurable plasmonics driven by epsilon-near-zero (ENZ) physics and time-varying media; (ii) alternative plasmonic materials— transition metal nitrides, transparent conducting oxides, and MXenes; (iii) quantum plasmonics and on-chip single-photon sources; (iv) solar energy conversion via plasmonic photocatalysis and interfacial solar-steam generation; (v) wearable surface-enhanced Raman spectroscopy (SERS), tip-enhanced Raman spectroscopy (TERS), and point-of-care diagnostics; (vi) thermal management and loss-mitigation in plasmonic integrated circuits; (vii) active tunable metasurfaces employing phase-change materials and hybrid Mie-plasmonic resonators; (viii) scalable nanofabrication and AI-assisted design; and (ix) van der Waals polaritonics and graphene plasmonics. For each frontier we identify key breakthroughs, examine the rigour of reported performance claims, highlight outstanding controversies, and propose directions for future work.

Keywords: Surface plasmon polaritons · Epsilon-near-zero materials · MXenes · Quantum plasmonics · Active metasurfaces · Solar energy conversion · SERS · Graphene plasmonics · Nanoimprint lithography

1 Introduction: The Paradigmatic Shift from Passive to Active Plasmonics

The discipline of plasmonics has traditionally been defined by the study of surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPR), phenomena arising from the collective oscillation of free electrons in metallic nanostructures coupled to electromagnetic fields [1, 2]. For decades, the field was dominated by noble metals—primarily gold and silver—owing to their high quality factors and resonant behaviour within the visible and near-infrared (NIR) spectra. However, the contemporary landscape of plasmonic research has undergone a radical transformation.

A clear transition is now evident from purely passive components to *active* and *reconfigurable* architectures whose optical response can be modulated post-fabrication through external stimuli such as electrical bias, thermal cycles, mechanical strain, or phase transitions [3, 4]. This evolution is fundamentally driven by the necessity to integrate plasmonic functionalities into practical electronic and photonic circuits. While conventional metallic nanostructures offer extraordinary light confinement, their fixed geometries limit utility in dynamic applications such as optical switching, adaptive lenses, and reconfigurable sensors [3, 4].

Modern theoretical frameworks encompass the *Active Metamaterials Roadmap*, which traces a trajectory towards "timetronics" and time-varying electromagnetic environments—where material properties are modulated not merely in space but also in time—enabling non-reciprocal light propagation and Floquet-engineered band structures [3, 5, 6]. It should be noted that the term "time crystal" in this photonic context refers to periodic temporal modulation of material parameters and should not be conflated with the distinct quantum-mechanical concept of discrete time-translation symmetry breaking studied in closed quantum systems [5].

1.1 The Drude Model and Dynamic Permittivity Engineering

The physics of the active-plasmonics shift rests on manipulation of the dielectric function. The classical Drude model describes the metallic response as

$$\varepsilon(\omega) = 1 - \frac{\omega_{\mathrm{p}}^2}{\omega^2 + i\gamma\omega}\,, \tag{1}$$

where ω_{p} is the plasma frequency and γ is the damping constant. Recent advances allow dynamic tuning of ω_{p} or γ through carrier injection or structural alteration of the lattice [2, 7]. This capability is critical for achieving *epsilon-near-zero* (ENZ) states, where the real part of the permittivity vanishes, leading to unprecedented nonlinearities and the ability to control the phase of light over deep-subwavelength distances [8, 9].

The ENZ regime is particularly compelling for active modulation. When $\mathrm{Re}[\varepsilon] \approx 0$, the phase velocity of light diverges while the group velocity remains finite and bounded by causality, enabling near-uniform phase accumulation across an entire thin film regardless of its physical geometry. This phenomenon, exploited in all-plasmonic Mach-Zehnder modulators and transparent conducting oxide (TCO) waveguide modulators, has enabled on-chip intensity modulation with extinction ratios up to ∼40 dB and energy consumptions below 25 fJ/bit [8, 10]. It is important to distinguish between the intrinsic *nonlinear* index change demonstrated by Alam et al. in ITO at ENZ [8] and the *modulator* performance achievable in a full device geometry, which additionally depends on overlap integrals, propagation length, and drive-circuit parasitics [10].

2 Advanced Material Systems Beyond Noble Metals

The dominance of gold and silver is being challenged by a new generation of "designer" plasmonic materials. Although noble metals possess high conductivity, they suffer from significant Ohmic losses, poor CMOS compatibility, and limited spectral tunability in the infrared (IR) range [1, 11]. The current research frontier is defined by transition metal nitrides (TMNs), transparent conducting oxides (TCOs), and two-dimensional (2D) materials such as MXenes [11–13].

Table 1: Summary of plasmonic material systems and their key properties. Q = quality factor; CMOS = complementary metal-oxide-semiconductor; NIR = near infrared; Mid-IR = mid infrared.

Material	Spectral Range	Key Advantage	Integration
Au, Ag	Visible	Highest Q factors	Poor (diffusion)
TiN, ZrN	Vis–NIR	Refractory, high thermal stability (inert atm.)	High (CMOS)
AZO, ITO	NIR	Electrical tunability (ENZ)	High (foundry)
$Ti_3C_2T_x$	NIR–Mid-IR	Hydrophilic, 2D morphology	High (solution)
Graphene	THz–Mid-IR	Extreme confinement ($\lambda/300$), gate-tunable (see Sect. 9)	Moderate (transfer)

2.1 Transition Metal Nitrides and Transparent Conducting Oxides

Transition metal nitrides (TMNs) such as titanium nitride (TiN) and zirconium nitride (ZrN) have emerged as *refractory* plasmonic materials [14]. These compounds offer high thermal stability and compatibility with standard semiconductor manufacturing, making them ideal for high-power applications and integration into silicon photonic chips [14]. Unlike gold, which melts at ∼1065 °C, TiN has a melting point above 2900 °C. However, it is important to note that maintaining plasmonic functionality at elevated temperatures requires inert or reducing atmospheres: in air, TiN begins to oxidise to optically insulating TiO_2 above ∼550 °C, which substantially degrades its plasmonic response [7, 14]. The thermal robustness claim therefore applies specifically to controlled-atmosphere applications such as spectrally selective emitters and high-temperature photonic chips packaged in inert environments.

In the NIR, transparent conducting oxides such as aluminium-doped zinc oxide (AZO) and indium tin oxide (ITO) provide a unique platform for electrical tunability [8, 15]. Hybrid plasmonic waveguides employing a thin AZO layer as the active medium exploit the ENZ effect: a low voltage (∼1 V) modulates the carrier density in the AZO layer to transiently reach the ENZ condition, facilitating phase and intensity modulation with high extinction ratios [15]. This mechanism outperforms traditional thermo-optic or electro-optic effects because it directly modulates the plasmonic resonance rather than inducing subtle refractive index changes in a bulk dielectric [8].

2.2 The Rise of MXenes in Sustainable Nanotechnology

MXenes—a diverse family of 2D transition metal carbides and nitrides—have been identified as "rising stars" in sustainable nanotechnology [13]. Discovered in 2011, MXenes such as $Ti_3C_2T_x$ exhibit high metallic conductivity, high breakdown voltages, and a hydrophilic nature that distinguishes them from other 2D materials such as graphene or transition metal dichalcogenides (TMDs) [12, 16]. Their aqueous processability enables fabrication of plasmonic structures via 3D printing and roll-to-roll manufacturing at room temperature [13, 16].

The optical properties of MXenes are highly tunable via surface termination chemistry. Single-layer Ti_3C_2 films exhibit strong broadband absorption extending into the near- and mid-infrared, arising from their high metallic conductivity and two-dimensional morphology [12, 16]. These properties have been exploited for photothermal cancer therapy and solar-driven desalination [13]. Recent studies have successfully integrated MXenes as alternative plasmonic coatings in one-dimensional photonic crystal platforms to support Tamm plasmon polaritons (TPP) [17], demonstrating high optical asymmetry and large high-to-low reflection ratios with minimal dielectric layers. A critical caveat is that MXene films are susceptible to oxidation in aqueous environments and under prolonged illumination, which degrades both electrical conductivity and optical response [13]; encapsulation strategies are an active research frontier.

3 Quantum Plasmonics and On-Chip Single-Photon Generation

The intersection of plasmonics and quantum optics—*quantum plasmonics*—exploits deep-subwavelength confinement to enhance quantum light-matter interactions.

This field underpins scalable quantum communication networks and photonic quantum computing [18, 19].

3.1 Deterministic Single-Photon Sources

A critical requirement for quantum systems is a reliable *single-photon emitter* (SPE) that generates precisely one photon per excitation cycle with high purity and indistinguishability [19]. While traditional methods were probabilistic—such as spontaneous parametric down-conversion (SPDC)—recent breakthroughs focus on deterministic sources using semiconductor quantum dots (QDs) [18, 19].

Table 2: Benchmark comparison of on-demand single-photon source technologies. QD = quantum dot; SPDC = spontaneous parametric down-conversion; PIC = photonic integrated circuit; MIR = mid-infrared; QKD = quantum key distribution.

Technology	Mechanism	Temperature
QD Circular Bragg	Shaped laser pulse	Cryogenic
Rare-Earth Ion Fibre	Nd^{3+} excitation	Room temp.
Phonon-Assisted	Heralded	Intermediate
SPDC Metasurface	$\chi^{(2)}$ nonlinear	Room temp.

Advances in nanophotonic engineering have redefined the performance limits of single-photon sources. Integration with circular Bragg resonators (CBRs) and photonic crystal (PC) cavities enhances photon extraction efficiency while maintaining high-fidelity polarisation entanglement [18, 19]. A recent landmark demonstration produced a chip that reliably emits single photons on demand by coupling quantum dots to carefully shaped laser pulses [20]. This advance is pivotal for quantum key distribution (QKD), as it eliminates the multi-photon "replication" risk inherent in probabilistic sources [20]. Simultaneously, researchers have demonstrated a room-temperature scheme for generating single photons from individual rare-earth ions in optical environments, offering low-loss insertion into existing telecommunications networks [21].

3.2 Entanglement Mechanics and High-Fidelity Quantum State Tomography

Bipartite entanglement in nanophotonic systems is achieved through the Hong-Ou-Mandel (HOM) effect, where two SPPs interfere on a plasmonic beamsplitter [22]. Quantum state tomography of these systems has confirmed that plasmonic platforms can sustain quantum interference despite their inherently dissipative nature [22]. The preservation of energy-time entanglement in plasmon-assisted light transmission has been verified experimentally, with measured concurrence values proving plasmonic coherence across multiple scattering events [23].

The shift from the traditional biexciton-exciton (XX-X) cascade to resonant or phonon-assisted single-excitation schemes represents a major practical advance [19]. In the cascade model, sequential two-photon emission from the biexciton through the intermediate exciton state is the dominant decay channel; however, fine-structure splitting and re-excitation between cascade photons impose constraints on entanglement fidelity. Modern resonant excitation techniques suppress these effects, achieving near-transform-limited photons with indistinguishabilities exceeding 98% and brightnesses above 57% directly into a single mode [18, 19].

Critical perspective. The plasmonic contribution to quantum photonic systems carries a fundamental tension: the very sub-wavelength confinement that accelerates spontaneous emission (Purcell factor $F_P > 10^3$ in gap modes) also introduces Ohmic absorption that reduces the fraction of photons radiated into useful modes (the β-factor). At room temperature, pure dephasing in solid-state emitters further limits single-photon purity. Most demonstrations of quantum interference in plasmonic circuits still require cryogenic cooling [22], and a room-temperature on-chip plasmonic single-photon source with simultaneously high purity, brightness, and indistinguishability remains an open challenge.

4 Solar Energy Conversion and Environmental Remediation

Plasmonic nanomaterials have become indispensable in solar energy harvesting owing to their LSPR-driven broadband sunlight absorption and efficient photothermal conversion [24].

4.1 Enhanced Photocatalytic Hydrogen Production

Green hydrogen production through photocatalytic water splitting is a flagship application of plasmonic composite systems. By coupling plasmonic nanoparticles (e.g., Au, Pd) with wide-bandgap semiconductors such as TiO_2 and ZnO, researchers have surmounted the thermodynamic barriers of traditional photocatalysis [24, 25]. Hot-electron injection from the metal nanoparticle into the conduction band of the semiconductor drives redox reactions under sub-bandgap illumination, dramatically extending the spectral utilisation window [25].

A notable mechanism involves *hot-electron injection* and interface passivation in Au/ZnO heterojunctions. Cushing et al. demonstrated that plasmon resonance energy transfer (PRET) from gold nanoparticles drives below-bandgap photocatalysis in ZnO, with the charge-transfer cross-section dependent on spectral overlap between the Au LSPR and the ZnO absorption edge [25]. The strategy highlights the importance of energy-level engineering at metal-semiconductor interfaces in plasmonic catalysis.

Critical perspective. A persistent controversy in plasmonic photocatalysis concerns whether the apparent rate enhancements arise from genuine hot-carrier injection or simply from local heating [24]. Distinguishing between the two mechanisms experimentally requires careful control of substrate temperature and use of isotopically labelled reactants or ultrafast spectroscopy. Many reported enhancement factors in the literature have been obtained under continuous-wave illumination where thermal effects are difficult to exclude, casting doubt on the claimed quantum efficiency values.

4.2 Solar-Driven Interfacial Evaporation and Desalination

Interfacial evaporators exploiting plasmonic materials have achieved conversion efficiencies and evaporation rates that were previously considered unattainable [26]. Table 3 benchmarks representative systems.

Table 3: Performance of plasmonic interfacial evaporator systems under 1-sun illumination (1 kW/m^2). Efficiencies are defined as $\eta = \dot{m}\Delta H_{\mathrm{vap}}/P_{\mathrm{solar}}$; values approaching or exceeding 100% indicate harvesting of environmental enthalpy (conduction/convection) in addition to solar input.

System	Efficiency (%)	Rate ($\mathbf{kg\,m^{-2}h^{-1}}$)
Al NP 3D Assembly	∼90	∼1.0
TiN Nanocomposites	≤99	1.76
Pd/Semiconductor	>89	>1.5
MXene/Au@$Cu_{2-x}S$	96.1	2.023

The three-dimensional aluminium nanoparticle assembly reported by Zhou et al. [26] pioneered the concept of plasmon-driven interfacial steam generation, achieving broadband solar absorption through coupled dipolar resonances across the visible spectrum. Subsequent work combining multiple plasmonic media (MXene, gold, and $Cu_{2-x}S$) has pushed light-to-heat conversion efficiencies above 96%, with evaporation rates of $\sim$2 $\mathrm{kg\,m^{-2}\,h^{-1}}$, demonstrating the synergy achievable through plasmonic heterostructures [16].

Critical perspective. Evaporation rates exceeding $\sim$1.44 $\mathrm{kg\,m^{-2}\,h^{-1}}$ under 1-sun illumination imply that the system harvests thermal energy from the ambient environment in addition to the solar flux, since the water vaporisation enthalpy ($\approx$ 2.45 MJ/kg) combined with 1 kW/m^2 yields a thermodynamic maximum of $\approx$ 1.47 $\mathrm{kg\,m^{-2}\,h^{-1}}$ at 100% efficiency. While such "super-efficiency" is physically permissible if the evaporator acts as a heat pump drawing from environmental enthalpy, comparisons between systems are meaningful only when the same efficiency accounting convention is applied [26]. Several recent studies have adopted non-standard definitions—dividing the evaporation enthalpy flux by the *absorbed* (rather than incident) solar flux—which can significantly inflate reported efficiency numbers.

5 Wearable Sensors and Point-of-Care Diagnostics

Plasmonic sensing—in particular surface-enhanced Raman spectroscopy (SERS)—has transitioned from specialised laboratory equipment to wearable healthcare solutions, leveraging nanoscale light-matter interactions to detect electrolytes, metabolites, hormones, and proteins in sweat or interstitial fluid [1, 27].

5.1 Integrated SERS and Wearable Platforms

Wearable plasmonic sensors are being embedded into smart contact lenses for continuous monitoring of intraocular pressure and glucose levels [28]. These platforms combine material innovations with machine-learning algorithms to interpret complex spectral data, improving specificity and reducing false-positive rates [28].

SERS substrates can be engineered to provide average electromagnetic field-enhancement factors of order 10^7–10^8, sufficient for single-molecule sensitivity at "hot spots" where the local enhancement can reach 10^{10}–10^{11} [27]. A critical distinction drawn by Le Ru et al. is that single-molecule SERS typically occurs at a tiny fraction of active sites on an otherwise moderate substrate; quoting only the single-molecule enhancement factor without context of hot-spot density can be misleading for quantitative sensor design [27]. Systematic characterisation of the axial SERS response reveals a Lorentzian signal profile along the optical axis, underlining the importance of focus precision for reproducible wearable measurements [29].

5.2 MXene-Based Biosensing

MXenes have emerged as superior substrates for electrochemical biosensors due to their large surface area, high electrical conductivity, and abundant surface functional groups that facilitate target capture [13]. The excellent electrochemical performance of $Ti_3C_2T_x$ was established through cation intercalation studies demonstrating high volumetric capacitance [30], properties equally valuable in sensing applications.

Antibody-functionalised $Ti_3C_2T_x$ nanosheets have been deployed for point-of-care detection of biomarkers at clinically relevant concentrations critical for screening of nutrient deficiencies and toxicity in resource-limited healthcare settings [16, 28]. The combination of MXene substrates with plasmonic enhancement further amplifies signal-to-noise ratios, promising multiplex sensing of biomarkers from a single micro-volume sample [13].

5.3 Tip-Enhanced Raman Spectroscopy

Tip-enhanced Raman spectroscopy (TERS) bridges the gap between diffraction- limited far-field SERS and true nanoscale chemical imaging. A metallic scanning-probe tip acts as a single plasmonic antenna, concentrating the excitation field into a sub-10 nm apex volume. Zhang et al. demonstrated sub-nanometre spatial resolution TERS on a single porphyrin molecule, resolving individual chemical bonds with an effective electromagnetic enhancement exceeding 10^9 [31]. This achievement established TERS as the de facto tool for single-molecule vibrational spectroscopy at surfaces, with direct implications for heterogeneous catalysis monitoring, DNA sequencing, and two-dimensional material defect characterisation. The main practical limitations are tip reproducibility, thermal drift in the tip-sample junction, and the requirement for ultra-low-vibration environments that currently restrict TERS to specialist laboratories [29, 31].

6 Thermal Management and Loss Mitigation in Integrated Circuits

As semiconductor devices approach kilowatt-class thermal design power—particularly in AI accelerators and three-dimensional integrated circuits (3D ICs)—thermal management has become the primary bottleneck for system performance [32, 33]. Plasmonic integrated circuits, operating at high speeds and subject to Ohmic losses, are especially susceptible [2, 10].

6.1 Cooling Inside the Silicon and Microfluidic Approaches

The industry is shifting toward "cooling inside the silicon," where microchannels are etched directly into silicon substrates to act as a "circulatory system" for coolant [32]. This approach routes coolant to within micrometres of active transistors or plasmonic modulators, often utilising two-phase (liquid-to-vapour) cooling to dramatically reduce thermal resistance [32].

Table 4: Recent thermal management advances relevant to plasmonic integrated circuits. TIM = thermal interface material; BAs = boron arsenide.

Technology	Key Metric	Status
Microfluidic direct-to-chip	Kilowatt-class extraction	Emerging
Graphene TIM	>10× greases	Active
Boron Arsenide	1300 $\mathrm{W\,m^{-1}K^{-1}}$	R&D
Solid-State Transistor	Tunable heat flow	Experimental

Materials with ultrahigh thermal conductivity, such as boron arsenide (BAs, ≈1300 $\mathrm{W\,m^{-1}K^{-1}}$ in bulk single crystals) and boron phosphide, have set new conductivity benchmarks [34]. These values are comparable to diamond and are achieved through a bunching of acoustic phonon branches that suppresses Umklapp scattering [34]. Graphene-based thermal interface materials provide more than a tenfold improvement over conventional thermal greases, accelerating heat spreading in densely integrated photonic chips [33]. Reduction of thermal boundary resistance (TBR) at heterointerfaces remains a key challenge for all high-density photonic and electronic circuits.

6.2 Compensating Losses via Optical Gain Integration

To address the intrinsic Ohmic losses of plasmonic systems, a powerful strategy employs optical gain media in the near field of plasmonic structures. De Leon and Berini demonstrated amplification of long-range SPPs by embedding the plasmonic waveguide in a dipolar gain medium, achieving net signal amplification that compensates propagation losses [35]. This constitutes a viable pathway for loss-sensitive applications in photonic circuits and sub-diffraction-limit imaging [35]. The all-plasmonic Mach-Zehnder modulator reported by Haffner et al. is a complementary demonstration that accepts Ohmic loss as a design variable, achieving record footprint ($< 10\ \mu$m active length) by trading propagation distance for extreme field confinement [10]. Combined with recent proposals for excitation at complex frequencies to temporarily overcome passive loss bounds, these techniques motivate continued exploration of gain-plasmonic hybrid architectures.

7 Active Tunable Metasurfaces and Reconfigurable Optics

Real-time optical reconfigurability defines the next generation of multifunctional optoelectronic devices. Active metasurfaces employ electrical modulation, mechanical strain, and thermal phase transitions to dynamically tailor amplitude, phase, and polarisation of light [3, 4, 36].

7.1 Vanadium Dioxide and Phase-Change Chalcogenides

Vanadium dioxide (VO_2) is a volatile phase-change material (PCM) that undergoes a metal-insulator transition at 68 °C, switching from a monoclinic (insulating) to a rutile (metallic) phase [4, 37]. Kats et al. demonstrated that a sub-wavelength VO_2 film can provide thermally switchable, near-unity absorption across the mid-infrared, establishing a benchmark for ultra-thin tunable perfect absorbers [37]. Upon thermally inducing the phase transition, absorption resonances can be spectrally tuned across tens of nanometres [37]. An important practical limitation is the pronounced thermal hysteresis of VO_2: the insulator-to-metal transition on heating occurs at ≈68 °C while the reverse metal-to-insulator transition on cooling occurs at ≈58 °C, producing a ∼10 °C hysteresis window that complicates precise analog reconfigurability [4].

Non-volatile PCMs such as GeSbTe (GST) and GeSbSeTe (GSST) retain their phase state after the initial stimulus is removed, making them ideal for programmable photonics and optical memory [36]. Wang et al. demonstrated optically reconfigurable metasurfaces and photonic devices using GST phase transitions, achieving on-demand pixel-wise control of amplitude and phase, enabling varifocal lensing and dynamic holography [3, 36]. A key challenge for non-volatile PCMs is write endurance: GST undergoes material redistribution and void formation after 10^4–10^6 switching cycles, whereas GSST offers improved optical contrast in the telecommunications C-band and better film stability but with reduced refractive index contrast relative to GST [36].

7.2 Hybrid Dielectric-Plasmonic Resonators

Dielectric nanoresonators are preferred over metallic counterparts for their lower energy dissipation, but suffer from less confined resonances and lower absolute field enhancement [38]. Hybrid Mie-plasmonic (MP) systems resolve this trade-off by combining the low losses of Mie-resonant dielectrics with the strong field confinement of plasmonics [38].

Supercavity modes with record-high quality factors arise from the coupling of Mie resonances with Fabry-Pérot-plasmonic modes near higher-order anapole conditions—a prediction confirmed experimentally by Rybin et al. with $Q > 260$ in silicon nanodiscs [39]. These high-Q resonances dramatically enhance nonlinear optical processes: in coupled dielectric-semiconductor nanoresonators, second-harmonic generation (SHG) signal intensities have been boosted by over an order of magnitude, a milestone for miniaturised frequency converters and entangled photon pair sources [40].

8 Scalable Nanofabrication and Industrial Adoption

The translation of plasmonic research from laboratory to manufacturing is enabled by scalable techniques such as nanoimprint lithography (NIL) and roll-to-roll (R2R) processing [41, 42].

8.1 Roll-to-Roll Nanoimprint Lithography

R2R-NIL provides continuous patterning of complex nanostructures on flexible polymers, glass foils, and thin films [42, 43]. UV-curing imprint engines can replicate gratings, metasurfaces, and diffractive optical elements with sub-100 nm resolution [43]. Achievable speeds span a wide range depending on feature size: coarse structures (>200 nm pitch) can be replicated at tens of metres per minute, whereas sub-100 nm features impose strict resist-flow and mould-filling constraints that typically limit throughput to below ∼1 m/min to maintain dimensional fidelity [42, 43]. This technology is currently being adopted for augmented and virtual reality (AR/VR) waveguide manufacturing, which demands high refractive indices, low surface roughness, and precise slanted or blazed grating geometries [42].

Table 5: Comparison of patterning technologies for plasmonic nanostructure manufacturing. EBL = electron-beam lithography.

Metric	R2R-NIL	R2P-NIL	EBL
Speed (m/min)	0.005–50	Medium	Slow
Throughput (m^2/yr)	1.5×10^4–4×10^6	Medium	Low
Resolution	<100 nm	<100 nm	<10 nm
Substrate	Flexible	Rigid	Wafer

Template stripping—a technique reported by Nagpal et al. for producing ultrasmooth patterned metal films with surface roughness below 0.5 nm [44]—yields plasmonic substrates whose low-roughness interfaces reduce radiation scattering losses and produce sharper resonances than conventional lift-off or etch processes. This approach has since been adapted to flexible and stretchable substrates, where gold nanohole arrays and pyramid-tip antennas on elastomeric films exhibit optical resonances that can be mechanically tuned by stretching the underlying film [44]. This mechanical tunability adds a new degree of freedom unavailable to rigid plasmonic substrates.

8.2 Additive Manufacturing and AI-Enhanced Design

The integration of plasmonic materials—particularly MXenes—into 3D printing workflows enables fabrication of intricate architectures for energy storage, electromagnetic shielding, and electronics [13, 16]. Artificial intelligence (AI) and machine-learning (ML) algorithms are now routinely employed to design and optimise plasmonic metasurfaces, predicting transmission and phase responses with low mean-squared errors across the visible and NIR spectral ranges [45]. Inverse design via deep neural networks and generative adversarial models dramatically reduces the design cycle time from weeks to minutes [45], democratising high-performance metasurface development beyond specialist computational electromagnetics groups.

9 Van der Waals Polaritonics and Graphene Plasmonics

A major research frontier that was not adequately represented in earlier plasmonics reviews is the broader family of *polaritons* in two-dimensional and van der Waals (vdW) materials, most notably graphene plasmons and phonon-polaritons in hexagonal boron nitride (hBN). These systems extend plasmonic confinement far beyond what noble metals can achieve and operate in spectral regimes inaccessible to conventional metal nanostructures.

9.1 Graphene Plasmons

Graphene supports transverse magnetic (TM) plasmons whose wavelength is compressed to $\lambda/100$–$\lambda/300$ of the free-space value in the terahertz and mid-infrared spectral ranges [46]. Unlike metal plasmons—whose carrier density is fixed by crystal stoichiometry—graphene plasmons are continuously gate-tunable: electrostatic gating shifts the Fermi energy by hundreds of meV, shifting the plasmon resonance frequency across the entire mid-infrared and dramatically changing its amplitude [46]. Near-field scanning optical microscopy measurements have directly imaged graphene plasmon polaritons propagating as standing waves from edges and defects, confirming the predicted extreme confinement [46].

The main limitation of graphene plasmons at room temperature is finite momentum lifetime: impurity scattering and phonon emission in the SiO_2 substrate reduce carrier mean free paths and hence plasmon propagation lengths. Encapsulation with hBN, which provides an atomically flat, phonon-free surface, can increase carrier mobilities to $>100\,000$ cm^{2}V^{-1}s^{-1} and plasmon quality factors to >50 at room temperature [46].

9.2 Phonon-Polaritons in Hexagonal Boron Nitride

Hexagonal boron nitride (hBN) supports hyperbolic phonon-polaritons (HPhPs) within two Reststrahlen bands where $\mathrm{Re}[\varepsilon_x]$ and $\mathrm{Re}[\varepsilon_z]$ have opposite signs, yielding an isofrequency surface that is hyperbolic in k-space [47]. These modes propagate with extreme spatial confinement ($\lambda_p/\lambda_0 \sim 1/60$) and quality factors exceeding 40 at room temperature—more than an order of magnitude higher than noble-metal plasmons at equivalent wavelengths [47]. The hyperbolic dispersion also enables sub-diffractional imaging and nanoscale heat transfer exceeding Planck's blackbody limit by orders of magnitude [47].

A critically important distinction from metal plasmonics is that HPhPs in hBN are supported by phonon motion rather than free-electron oscillation; consequently they are confined to the two Reststrahlen bands (≈760–825 cm^{-1} and ≈1370–1610 cm^{-1}) and cannot be continuously spectrally tuned via gating. Hybridising hBN with graphene or other 2D conductors creates gate-tunable phonon-plasmon-polaritons that partially overcome this restriction.

9.3 Picocavity Effects and Quantum Corrections

As plasmonic nano-gaps are reduced below ∼1 nm, classical electromagnetic modelling based on bulk permittivities breaks down. Quantum corrections—including electron-density spill-out, non-local screening, and optical tunnelling—alter the resonance frequency and field enhancement relative to classical predictions [48]. Benz et al. demonstrated that individual atomic-scale protrusions in a sub-nanometre gap spontaneously form and dissolve thermally, creating ultraconfined "picocavities" with mode volumes of order $(1\ \text{Å})^3$ [48]. These picocavities can enhance vibrational spectroscopy signals by an additional factor of $\sim 10^3$ over a conventional nanogap, enabling observation of the Raman spectrum of a single chemical bond as a function of time. This work signals the entry of plasmonics into the quantum-mechanical regime even for room-temperature, classical metallic structures [48].

10 Outlook and Conclusions

Plasmonics stands at a pivotal juncture, but progress towards technological deployment has been uneven across the subfields reviewed here. Some areas—notably all-plasmonic modulators and SERS-based point-of-care diagnostics—have reached early commercial adoption, while others, particularly room-temperature quantum plasmonic sources and large-scale solar-steam systems, face challenges that have been understated in much of the recent literature. The following trends will likely define the next decade, alongside honest assessment of the remaining hurdles:

1. **Time-varying and active media.** The integration of ENZ materials and phase-change media with time-varying electromagnetic environments will unlock non-reciprocal optical devices and Floquet-engineered metamaterials [3, 6, 9]. The terminology "time crystals" in the photonic context should be distinguished from the quantum many-body phenomenon, and the transition from single-element demonstrations to integrated chips requires solving thermal cross-talk and drive-latency issues.
2. **Sustainable, CMOS-compatible materials.** The replacement of noble metals with TMNs, TCOs, and MXenes will enable mass production within existing foundry infrastructure [11, 14], but oxidation stability (TiN, MXenes) and CMOS contamination (ITO containing indium) remain open engineering challenges.
3. **Quantum advantage at room temperature.** Room-temperature single-photon emission from rare-earth ions [21], combined with plasmonic enhancement of light-matter coupling, is a compelling long-term goal; however, pure dephasing rates at 300 K remain orders of magnitude larger than at cryogenic temperatures.
4. **Convergence with AI.** Inverse design algorithms trained on large experimental datasets will enable discovery of new plasmonic geometries [45]; systematic dataset curation and uncertainty quantification are critical prerequisites to avoid overfit models.
5. **Van der Waals polaritonics.** Graphene plasmons and hBN phonon-polaritons provide confinement and quality factors inaccessible to metals [46, 47]; hybrid van der Waals/metallic structures represent the next frontier.
6. **Wearable and implantable photonics.** The mechanical flexibility of MXene and NIL-patterned substrates, combined with on-chip signal processing, will realise wearable plasmonic spectrometers [13, 28]; biocompatibility and sterilisation protocols for implantable variants require dedicated investigation.
7. **Thermal co-design.** As photonic integration densities rival microelectronics, thermal management must be a first-class design variable, necessitating co-simulation of optical, electrical, and thermal domains [32, 34].

In summary, the advances reviewed here demonstrate that plasmonics has broadened from a condensed matter physics curiosity into an enabling technology platform. However, a rigorous reading of the literature reveals that several widely cited performance claims require important caveats: evaporation efficiencies exceeding the solar thermodynamic limit rely on environmental enthalpy harvesting; TiN thermal robustness applies only in non-oxidising atmospheres; reported single-molecule SERS enhancement factors apply to a small fraction of hot-spot sites; and quantum-optical demonstrations in plasmonic circuits remain predominantly cryogenic. Addressing these nuances honestly is essential for the field to set realistic milestones and avoid overselling timelines for commercial deployment.

Declarations

Conflict of Interest The author declares no conflict of interest.

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References

[1] Stefan A. Maier. *Plasmonics: Fundamentals and Applications*. Springer, New York, 2007. ISBN 978-0-387-33150-8. doi: 10.1007/0-387-37825-1.

[2] Lukáš Novotny and Bert Hecht. *Principles of Nano-Optics*. Cambridge University Press, Cambridge, 2 edition, 2012. ISBN 978-1-107-00546-4. doi: 10.1017/CBO9780511794193.

[3] Amr M. Shaltout, Vladimir M. Shalaev, and Mark L. Brongersma. Spatiotemporal light control with active metasurfaces. *Science*, 364(6441):eaat3100, 2019. doi: 10.1126/science.aat3100.

[4] Hou-Tong Chen, Antoinette J. Taylor, and Nanfang Yu. A review of metasurfaces: physics and applications. *Reports on Progress in Physics*, 79(7):076401, 2016. doi: 10.1088/0034-4885/79/7/076401.

[5] Nikolay I. Zheludev and Yuri S. Kivshar. From metamaterials to metadevices. *Nature Materials*, 11 (11):917–924, 2012. doi: 10.1038/nmat3431.

[6] Nader Engheta. Pursuing near-zero response. *Science*, 340(6130):286–287, 2013. doi: 10.1126/science.1235589.

[7] Alexandra Boltasseva and Harry A. Atwater. Low-loss plasmonic metamaterials. *Science*, 331(6015): 290–291, 2011. doi: 10.1126/science.1198258.

[8] M. Zahirul Alam, Israel De Leon, and Robert W. Boyd. Large optical nonlinearity of indium tin oxide in its epsilon-near-zero region. *Science*, 352(6287): 795–797, 2016. doi: 10.1126/science.aae0330.

[9] Iván Liberal and Nader Engheta. Near-zero refractive index photonics. *Nature Photonics*, 11(3):149–158, 2017. doi: 10.1038/nphoton.2017.13.

[10] Christian Haffner, Wolfgang Heni, Yuriy Fedoryshyn, Jens Niegemann, Aram Melikyan, D. L. Elder, Benedikt Baeuerle, Yannick Salamin, Andreas Josten, Ulrich Koch, Claudia Hoessbacher, Florian Ducry, Lirong Juchli, Alexandros Emboras, David Hillerkuss, Michael Kohl, Larry R. Dalton, Christian Hafner, and Juerg Leuthold. All-plasmonic Mach–Zehnder modulator enabling optical high-speed communication at the microscale. *Nature Photonics*, 9 (8):525–528, 2015. doi: 10.1038/nphoton.2015.127.

[11] Gururaj V. Naik, Vladimir M. Shalaev, and Alexandra Boltasseva. Alternative plasmonic materials: Beyond gold and silver. *Advanced Materials*, 25(24): 3264–3294, 2013. doi: 10.1002/adma.201205076.

[12] Mohamed Naguib, Vadym N. Mochalin, Michel W. Barsoum, and Yury Gogotsi. 25th anniversary article: MXenes: A new family of two-dimensional materials. *Advanced Materials*, 26(7):992–1005, 2014. doi: 10.1002/adma.201304138.

[13] Yury Gogotsi and Babak Anasori. The rise of MXenes. *ACS Nano*, 13(8):8491–8494, 2019. doi: 10.1021/acsnano.9b06394.

[14] Urcan Guler, Vladimir M. Shalaev, and Alexandra Boltasseva. Nanoparticle plasmonics: going practical with transition metal nitrides. *Materials Today*, 18 (4):227–237, 2015. doi: 10.1016/j.mattod.2014.10.039.

[15] Eyal Feigenbaum, Kenneth Diest, and Harry A. Atwater. Unity-order index change in transparent conducting oxides at visible frequencies. *Nano Letters*, 10(6):2111–2116, 2010. doi: 10.1021/nl1006307.

[16] Babak Anasori, Maria R. Lukatskaya, and Yury Gogotsi. 2D metal carbides and nitrides (MXenes) for energy storage. *Nature Reviews Materials*, 2: 16098, 2017. doi: 10.1038/natrevmats.2016.98.

[17] M. Kaliteevski, I. Iorsh, S. Brand, R. A. Abram, J. M. Chamberlain, A. V. Kavokin, and I. A. Shelykh. Tamm plasmon-polaritons: Possible electromagnetic states at the interface of a metal and a dielectric Bragg mirror. *Physical Review B*, 76(16):165415, 2007. doi: 10.1103/PhysRevB.76.165415.

[18] Peter Lodahl, Sahand Mahmoodian, and Søren Stobbe. Interfacing single photons and single quantum dots with photonic nanostructures. *Reviews of Modern Physics*, 87(2):347–400, 2015. doi: 10.1103/RevModPhys.87.347.

[19] Pascale Senellart, Glenn Solomon, and Andrew White. High-performance semiconductor quantum-dot single-photon sources. *Nature Nanotechnology*, 12(11):1026–1039, 2017. doi: 10.1038/nnano.2017.218.

[20] Natasha Tomm, Alisa Javadi, Nadia Olympia Antoniadis, Daniel Najer, Matthias C. Löbl, Alexander R. Korsch, Ruediger Schott, Sascha R. Valentin, Andreas D. Wieck, Arne Ludwig, and Richard J. Warburton. A bright and fast source of coherent single photons. *Nature Nanotechnology*, 16(4):399–403, 2021. doi: 10.1038/s41565-020-00831-x.

[21] A. M. Dibos, M. Raha, C. M. Phenicie, and J. D. Thompson. Atomic source of single photons in the telecom band. *Physical Review Letters*, 120(24): 243601, 2018. doi: 10.1103/PhysRevLett.120.243601.

[22] Reinier W. Heeres, Leo P. Kouwenhoven, and Val Zwiller. Quantum interference in plasmonic circuits. *Nature Nanotechnology*, 8(10):719–722, 2013. doi: 10.1038/nnano.2013.178.

[23] S. Fasel, F. Robin, E. Moreno, D. Erni, N. Gisin, and H. Zbinden. Energy-time entanglement preservation in plasmon-assisted light transmission. *Physical Review Letters*, 94(11):110501, 2005. doi: 10.1103/PhysRevLett.94.110501.

[24] Suljo Linic, Phillip Christopher, and David B. Ingram. Plasmonic-metal nanostructures for efficient conversion of solar to chemical energy. *Nature Materials*, 10(12):911–921, 2011. doi: 10.1038/nmat3151.

[25] Scott K. Cushing, Jiuyang Li, Fanke Meng, Thomas R. Senty, Savan Suri, Mingjia Zhi, Mo Li, Alan D. Bristow, and Nianqiang Wu. Photocatalytic activity enhanced by plasmonic resonant energy transfer from metal to semiconductor. *Journal of the American Chemical Society*, 134(36):15033–15041, 2012. doi: 10.1021/ja305603t.

[26] Lin Zhou, Yingling Tan, Jiayu Wang, Wencai Xu, Ye Yuan, Wenshan Cai, Shining Zhu, and Jia Zhu. 3D self-assembly of aluminium nanoparticles for plasmon-enhanced solar desalination. *Nature Photonics*, 10(6):393–398, 2016. doi: 10.1038/nphoton.2016.75.

[27] E. C. Le Ru, E. Blackie, M. Meyer, and P. G. Etchegoin. Surface enhanced raman scattering enhancement factors: A comprehensive study. *Journal of Physical Chemistry C*, 111(37):13794–13803, 2007. doi: 10.1021/jp0687908.

[28] Jayoung Kim, Alan S. Campbell, Berta Estévez-Fernández de Ávila, and Joseph Wang. Wearable biosensors for healthcare monitoring. *Nature Biotechnology*, 37(4):389–406, 2019. doi: 10.1038/s41587-019-0045-y.

[29] Shun-Yeh Ding, Jian Yi, Jian-Feng Li, Bin Ren, De-Yin Wu, Rajapandiyan Panneerselvam, and Zhong-Qun Tian. Nanostructure-based plasmon-enhanced Raman spectroscopy for surface analysis of materials. *Nature Reviews Materials*, 1:16021, 2016. doi: 10.1038/natrevmats.2016.21.

[30] Maria R. Lukatskaya, Olha Mashtalir, Chang E. Ren, Yohan Dall'Agnese, Patrick Rozier, Pierre-Louis Taberna, Mohamed Naguib, Patrice Simon, Michel W. Barsoum, and Yury Gogotsi. Cation intercalation and high volumetric capacitance of two-dimensional titanium carbide. *Science*, 341(6153):1502–1505, 2013. doi: 10.1126/science.1241488.

[31] Rui Zhang, Yang Zhang, Z. C. Dong, S. Jiang, C. Zhang, L. G. Chen, L. Zhang, Y. Liao, J. Aizpurua, Y. Luo, J. L. Yang, and J. G. Hou. Chemical mapping of a single molecule by plasmon-enhanced Raman scattering. *Nature*, 498(7452):82–86, 2013. doi: 10.1038/nature12151.

[32] Suresh V. Garimella, Amy S. Fleischer, Jayathi Y. Murthy, Ali Keshavarzi, Ravi Prasher, Chetan Patel, Sushil H. Bhavnani, Rama Venkatasubramanian, Ravi Mahajan, Yogendra Joshi, Bahgat Sammakia, Bruce A. Myers, Len Chorosinski, Martine Baelmans, Pramod Sathyamurthy, and Puneet E. Bhatt. Thermal challenges in next-generation electronic systems. *IEEE Transactions on Components and Packaging Technologies*, 31(4):801–815, 2008. doi: 10.1109/TCAPT.2008.2001197.

[33] Alexander A. Balandin. Thermal properties of graphene and nanostructured carbon materials. *Nature Materials*, 10(8):569–581, 2011. doi: 10.1038/nmat3064.

[34] Fei Tian, Bai Song, Xi Chen, Navaneetha K. Ravichandran, Yinchuan Lv, Ke Chen, Sean Sullivan, Jaehyun Kim, Yaoying Zhou, Te-Huan Liu, Geethal Amila Gamage Goni, Zhiwei Ding, Jingying Sun, Geethal Amila Gamage, Hangbo Sun, Hasan Ziyaee, Shanyuan Huyan, Lijun Deng, Jun Zhou, Aaron J. Schmidt, Shuo Chen, Ching-Wu Chu, Po-Yen Huang, David Broido, Li Shi, Gang Chen, and Zhifeng Ren. Unusual high thermal conductivity in boron arsenide bulk crystals. *Science*, 361(6402):582–585, 2018. doi: 10.1126/science.aat7932.

[35] Israel De Leon and Pierre Berini. Amplification of long-range surface plasmons by a dipolar gain medium. *Nature Photonics*, 4(6):382–387, 2010. doi: 10.1038/nphoton.2010.37.

[36] Qian Wang, Edward T. F. Rogers, Behrad Gholipour, Chih-Ming Wang, Guanghui Yuan, Jinghua Teng, and Nikolay I. Zheludev. Optically reconfigurable metasurfaces and photonic devices based on phase change materials. *Nature Photonics*, 10(1):60–65, 2016. doi: 10.1038/nphoton.2015.247.

[37] Mikhail A. Kats, Deepika Sharma, Jiao Lin, Patrice Genevet, Romain Blanchard, Zheng Yang, Mumtaz M. Qazilbash, Dimitri N. Basov, Shriram Ramanathan, and Federico Capasso. Ultra-thin perfect absorber employing a tunable phase change material. *Applied Physics Letters*, 101(22):221101, 2012. doi: 10.1063/1.4767646.

[38] Vincenzo Giannini, Antonio I. Fernández-Domínguez, Susannah C. Heck, and Stefan A. Maier. Plasmonic nanoantennas: Fundamentals and their use in controlling the radiative properties of nanoemitters. *Chemical Reviews*, 111(6):3888–3912, 2011. doi: 10.1021/cr1002672.

[39] Mikhail V. Rybin, Kirill L. Koshelev, Zarina F. Sadrieva, Kirill B. Samusev, Andrey A. Bogdanov, Mikhail F. Limonov, and Yuri S. Kivshar. High-q supercavity modes in subwavelength dielectric resonators. *Physical Review Letters*, 119(24):243901, 2017. doi: 10.1103/PhysRevLett.119.243901.

[40] Kirill Koshelev, Sergey Kruk, Elizaveta Melik-Gaykazyan, Jae-Hyuck Choi, Andrey Bogdanov, Hong-Gyu Park, and Yuri Kivshar. Subwavelength dielectric resonators for nonlinear nanophotonics. *Science*, 367(6475):288–292, 2020. doi: 10.1126/science.aaz5835.

[41] S. V. Sreenivasan. Nanoscale manufacturing enabled by imprint lithography. *MRS Bulletin*, 33(9):854–863, 2008. doi: 10.1557/mrs2008.179.

[42] N. Kooy, K. Mohamed, L. T. Pin, and O. S. Guan. A review of roll-to-roll nanoimprint lithography. *Nanoscale Research Letters*, 9(1):320, 2014. doi: 10.1186/1556-276X-9-320.

[43] Se Hyun Ahn and L. Jay Guo. High-speed roll-to-roll nanoimprint lithography on flexible plastic substrates. *Advanced Materials*, 20(11):2044–2049, 2008. doi: 10.1002/adma.200702650.

[44] Prashant Nagpal, Nathan C. Lindquist, Sang-Hyun Oh, and David J. Norris. Ultrasmooth patterned metals for plasmonics and metamaterials. *Science*, 325 (5940):594–597, 2009. doi: 10.1126/science.1174655.

[45] Itzik Malkiel, Michael Mrejen, Achiya Nagler, Uri Arieli, Lior Wolf, and Haim Suchowski. Plasmonic nanostructure design and characterization via Deep Learning. *Light: Science & Applications*, 7:60, 2018. doi: 10.1038/s41377-018-0060-7.

[46] Alexander N. Grigorenko, Marco Polini, and K. S. Novoselov. Graphene plasmonics. *Nature Photonics*, 6(11):749–758, 2012. doi: 10.1038/nphoton.2012.262.

[47] Joshua D. Caldwell, Andrey V. Kretinin, Yiguo Chen, Vincenzo Giannini, Michael M. Fogler, Yan Francescato, Chase T. Ellis, Joseph G. Tischler, Colin R. Woods, Alexander J. Giles, Minghui Hong, Kenji Watanabe, Takashi Taniguchi, Stefan A. Maier, and Kostya S. Novoselov. Sub-diffractional volume-confined polaritons in the natural hyperbolic material hexagonal boron nitride. *Nature Communications*, 5:5221, 2014. doi: 10.1038/ncomms6221.

[48] Felix Benz, Mikolaj K. Schmidt, Alexander Dreismann, Rohit Chikkaraddy, Yao Zhang, Angela Demetriadou, Cloudy Carnegie, Hamid Ohadi, Bart de Nijs, Ruben Esteban, Javier Aizpurua, and Jeremy J. Baumberg. Single-molecule optomechanics in “picocavities”. *Science*, 354(6313):726–729, 2016. doi: 10.1126/science.aah5243.